[INCH-POUND]
A-A-59140
24 November 1997

COMMERCIAL ITEM DESCRIPTION

TEST SET, RADIO FREQUENCY POWER
(POWER METER, DIGITAL)
(50 MHz to 26.5 GHz)

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes a digital, single channel, power meter with sensors, used for measuring continuous wave (CW) microwave and radio frequency (RF) power over selected frequencies. This CID is meant as a minimum requirement for the power meter in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

2. SALIENT CHARACTERISTICS.

2.1. Equipment covered by this CID shall be portable commercial-off-the-shelf (COTS) equipment and may be modified to the extent necessary to meet the following description. The equipment shall be Class 3, in accordance with MIL-PRF-28800.

2.2. Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800.

2.3. Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 110/120 VAC (±10 %) at line frequencies of 50 Hz to 60 Hz and 380 to 420 Hz and 220/240 VAC (±10 %) at line frequencies of 50 Hz to 60 Hz. Power consumption shall not exceed 70 VA, 40 watts. Transient state conditions shall meet MIL-PRF-28800.

2.4. Temperature Requirements.

2.4.1. Operating Temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0°C to +50°C.

2.4.2. Non-operating Temperature. The equipment shall meet its performance and accuracy requirements after being in a non-operating (storage) environment of -10°C to +60°C.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Technology & Industrial Services Division, SA-ALC/TILDD, Bldg. 171, Post C-12, 485 Quentin Roosevelt Rd., Kelly AFB TX 78241-6425.

AMSC N/A FSC 6625

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:04Z
Check the source to verify that this is the current version before use.

2.5. Mean Time Between Failures (MTBF). The equipment MTBF shall be 2000 hours. The lower test MTBF shall be 2000 hours and the upper test MTBF shall be 4000 hours.

2.6. Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling with thermal cycling being accomplished in the sample testing phase.

2.7. Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. Adjustments shall be provided by variable value components which are adjustable by the use of simple means. Calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments shall be accessible without removal of the instrument case or modules wherever possible. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours.

2.8. Performance Characteristics.

2.8.1. Frequency Range. The minimum frequency range of the power meter with sensors shall be 50.0 MHz to 26.5 GHz.

2.8.2. Power Range. The minimum power range of the meter shall be -60 dBm to +20 dBm.

2.8.3. Front Panel Controls. Controls for the power meter shall be on the front-panel.

2.8.3.1. The power meter shall have the selection of either linear (mW) or logarithmic (dBm) modes for the display of the power measurements.

2.8.4. Display. The power meter shall have a display with a minimum of 4 digits for all functions and data.

2.8.4.1. The display shall also have indicators for measurement mode, display units (watts, dBm, dB), calibration factor, error conditions, and offset value.

2.8.5. Power Meter Accuracy.

2.8.5.1. The accuracy shall be +/-0.02 dB full scale in logarithmic mode and +/-0.5% in linear mode.

2.8.6. Power Reference Calibrator. The power reference source shall be an internal 50 MHz oscillator at 1.00 mW with a Type N connector on the front panel. The power reference shall be factory set to +/-0.7% traceable to the US National Institute of Standards and Technology.

2.8.6.1. Power Reference Accuracy. The power reference accuracy shall be ± 2.5 % at 1.0 mW for one year.

2.8.6.2. Power Reference Standing Wave Ratio (SWR). The SWR shall be less than 1.05.

2.9. Calibration Factor. The calibration factors shall be in steps of percent or dB. The calibration factors shall be stored in non-volatile memory. The preferred method is to enter the calibration factors by front panel.

2.10. CW Power Sensors. The CW power sensors shall have a minimum frequency range of 50 MHz to 26.5 GHz and a minimum power range of -60 dBm to +20 dBm.

2.10.1. Interconnecting Cable. The power meter shall include a cable for connecting the sensor to the meter. The minimum length of the cable shall be five (5) feet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:04Z
Check the source to verify that this is the current version before use.

2.10.2. RF Connection. The sensor RF interconnect shall be for a coaxial cable connector, Type N, K, APC-3.5 mm or 2.4 mm with a nominal impedance of 50 ohms.

2.10.3. Sensor SWR. The sensors shall have a maximum 1.50 SWR for the frequency and power specified in paragraph 2.10.

2.10.4. Sensor Calibration. The sensor calibration shall be automatic and self-calibrated to 1.00 mW.

2.11. Instrument Zeroing. The power meter shall have automatic zeroing to within ±0.5 % full scale using the highest sensitivity range.

2.12. Remote Operation. The power meter shall have an IEEE 488 interface for remote set up, control, and measurement operation.

2.13. Dimensions and Weight.

2.13.1. Dimensions. The dimensions for the power meter shall be approximately 4.0 inches high x 8.5 inches wide x 18.0 inches deep.

2.13.2. Weight. The weight, including all accessories (excluding manuals) shall not exceed 20 lb.

2.14. Rackmount. Rackmount capability shall be available as an option for the equipment specified herein.

3. REGULATORY REQUIREMENTS.

3.1. In accordance with Section 23.043 of the Federal Acquisition Regulation, the government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

4. QUALITY ASSURANCE PROVISIONS.

4.1. Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5. PACKAGING.

5.1. Preservation, Packaging, Labeling, and Marking. Preservation, packaging, labeling, and marking shall be as specified in the contract or order.

6. NOTES

6.1. Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

6.2. Known Acceptable Products. The following product(s) are known to have previously met the requirements of this CID and need not resubmit bid samples or certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:04Z
Check the source to verify that this is the current version before use.

Manufacturer	Model	NSN
Hewlett Packard (CAGE 28480)	HP437B (power meter) with sensors HP8485A and HP8485D with sensor cable HP11730A	6625-01-356-9554 6625-01-178-7627 6625-01-326-6499 5995-01-388-5763
Boonton (CAGE 04901)	4220A-01 (power meter) with 51071 sensor	6625-01-434-8496YA None

6.3. Source of Documents.

6.3.1. Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Defense Documents Order Desk, 700 Robbins Ave., Bldg 4D, Philadelphia PA 19111-5094. Ph: 215-697-2179.

MILITARY INTEREST:

Custodians
Air Force - 99

Reviewer

Preparing Activity
Air Force - 82

CIVIL AGENCY
COORDINATING ACTIVITY

Project No. 6625-0852

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:04Z
Check the source to verify that this is the current version before use.